

Expected Goals (xG) Modelling in Football

A Mathematical Modelling Approach

ACM 40960 · Mathematical Modelling

Abstract. This project builds and studies a calibrated Expected Goals (xG) model for football using the StatsBomb open dataset. The work treats xG as a mathematical model rather than only a prediction task. The two main shot features, distance and angle, are derived from pitch geometry, the baseline logistic model is framed as a maximum likelihood optimisation problem, and the fitted model is then extended in three directions: a Monte Carlo match simulation based on the Poisson-binomial distribution, a Bayesian version that quantifies uncertainty, and a sensitivity analysis. A simple distance and angle model reaches an AUC of 0.75, and adding contextual features raises this to 0.78. Random Forest and XGBoost match the enhanced logistic model almost exactly, which shows that with these features the choice of algorithm matters far less than the information available. All models are well calibrated. The simulation recovers known real-world results, identifying Leicester City and Atlético Madrid as the largest over-performers of the 2015/16 season.

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1 Introduction

Expected Goals (xG) is a way of measuring how likely a shot is to end up as a goal. Each shot is given a value between 0 and 1, where a higher value means a better chance of scoring, based on features such as the distance from goal, the angle to goal, the body part used, and the amount of defensive pressure. Over the last decade xG has become a standard metric in football analytics and is now used widely by clubs, broadcasters, and researchers. Because goals are rare, a measure that estimates the quality of a shot rather than just counting goals gives a more stable picture of performance.

The aim of this project is not only to fit an accurate xG model but to treat it as a mathematical object. The shot geometry is derived from first principles, the fitting of the baseline model is framed as an optimisation problem, the model is extended to whole match outcomes through simulation, and the uncertainty in the predictions is quantified. The work is carried out entirely on open data so that it remains reproducible.

2 Literature Review

Early xG models were kept deliberately simple. Rathke (2017) showed that a logistic regression model based mainly on shot distance and angle could already give a reasonable estimate of goal probability, which set the basic template that most later work has built on. Much of the later research focused on adding more information and using more flexible methods. Lucey et al. (2015) showed that strategic and spatial features improved prediction beyond shot location alone, and Anzer and Bauer (2021) combined synchronised positional and event data to build one of the more detailed models in the field. Mead et al. (2023) argued that standard models leave out useful information such as player and team ability, and researchers have increasingly moved from logistic regression toward tree based ensemble methods such as Random Forest and gradient boosting.

A separate strand of work has focused on interpretability and on adjusting models for who takes the shot. Hewitt and Karakuş (2023) trained position adjusted models, and Scholtes and Karakuş (2024) used a Bayesian hierarchical approach to correct for player and position. A common use of xG is to judge finishing ability by comparing a player’s actual goals to their total xG. This use has been questioned: Davis and Robberechts (2024) found that a player’s overperformance is not stable from one season to the next, and that xG models can carry biases because stronger players and teams take more shots. Their work is a useful reminder that a finishing table should be read with care rather than treated as a clean measure of skill.

xG is also a natural building block for modelling whole match outcomes. A long line of work models the number of goals a team scores using Poisson based models, starting with Dixon and Coles (1997), and later placed in a Bayesian hierarchical framework by Baio and Blangiardo (2010). Because each shot already has a goal probability, the goals in a match can be treated as a sum of independent Bernoulli trials, which links the shot level model to a simulation of results at the match level. The present project draws these threads together on open data, with a particular emphasis on calibration and on the mathematical structure behind each step.

3 Data

The data source is the StatsBomb open dataset. In the open release only the 2015/16 season is fully covered for the major leagues, so the four full 2015/16 seasons of La Liga, the Premier League, Serie A, and Ligue 1 were combined into a single dataset. This gives a large and unbiased sample rather than the player-specific subsets available for other seasons. After removing penalties, which are always taken from the same spot and would distort a distance and angle

model, the dataset contains 37,488 shots, of which 3,569 were goals, an overall conversion rate of 9.5 percent. Each shot records its pitch coordinates, body part, technique, shot type, play pattern, whether the shooter was under pressure, and whether it was a first-time shot.

4 Shot Geometry

The pitch is 120 units long and 80 units wide. The goal being attacked lies on the line $x = 120$, with posts at $y = 36$ and $y = 44$, so the goal is $w = 8$ units wide and its centre is $C = (120, 40)$. Two features are derived from the raw location (x, y) .

The distance is the straight-line distance to the centre of the goal,

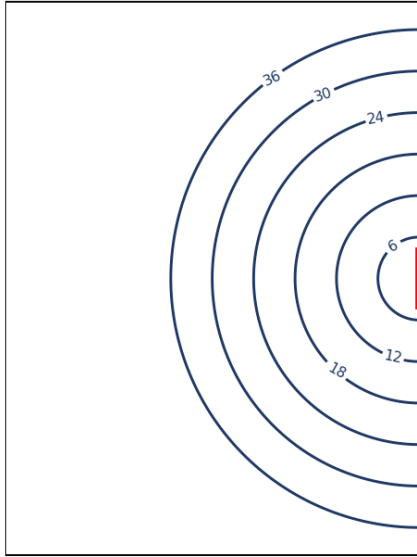
$$d = \sqrt{(120 - x)^2 + (40 - y)^2}. \quad (1)$$

The angle is the width of the goal as seen from the shot, that is, the angle θ subtended at $P = (x, y)$ by the two posts $A = (120, 36)$ and $B = (120, 44)$. With $p = |PA|$ and $q = |PB|$ the distances to each post, the law of cosines applied to triangle APB , whose side opposite θ is the goal of length $w = 8$, gives

$$\theta = \arccos\left(\frac{p^2 + q^2 - w^2}{2pq}\right). \quad (2)$$

Figure 1 shows lines of equal distance and equal angle on the pitch. Each equal-angle curve is a circular arc through both posts, a consequence of the inscribed angle theorem. Figure 2 confirms that both features carry genuine signal: conversion falls sharply with distance and rises with a wider angle, which justifies using d and θ as the inputs to the baseline model.

Iso-distance curves (units to goal centre)



Iso-angle curves (goal width seen, degrees)

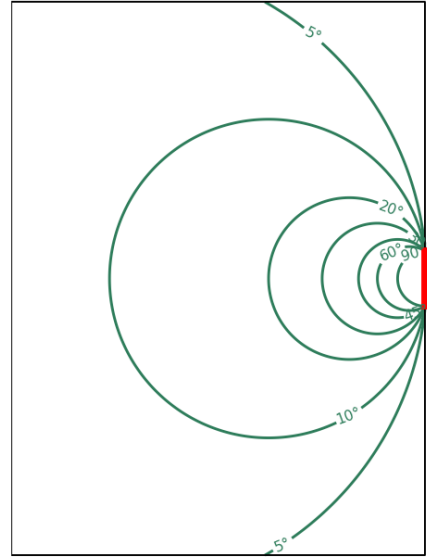


Figure 1: Lines of equal distance (left) and equal angle (right). The red segment is the goal.

5 Baseline Model: Logistic Regression

For each shot i with distance d_i and angle θ_i and outcome $y_i \in \{0, 1\}$, the probability of a goal is modelled as

$$p_i = \sigma(\eta_i) = \frac{1}{1 + e^{-\eta_i}}, \quad \eta_i = \beta_0 + \beta_1 d_i + \beta_2 \theta_i, \quad (3)$$

which is linear on the log-odds scale, $\log(p_i/(1 - p_i)) = \beta_0 + \beta_1 d_i + \beta_2 \theta_i$.

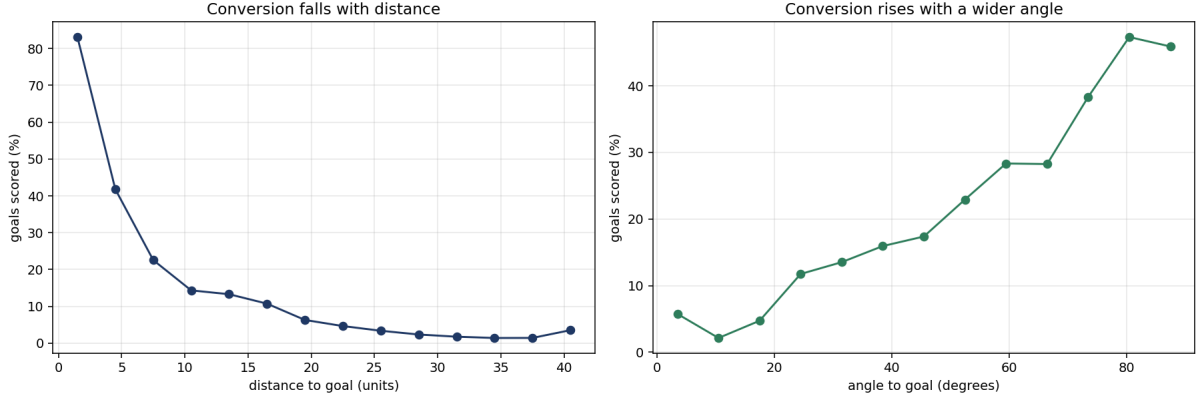


Figure 2: Empirical conversion rate against distance and angle, confirming both features are predictive.

Fitting as an optimisation problem. Each outcome is a Bernoulli trial, so the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (4)$$

and the maximum likelihood estimate is $\hat{\beta} = \arg \max_{\beta} \ell(\beta)$. Setting the gradient to zero gives the score equations $\mathbf{X}^T(\mathbf{y} - \mathbf{p}) = \mathbf{0}$, which have no closed-form solution and are solved numerically by Newton–Raphson. The log-likelihood is concave, so the solution is the unique global maximum.

The fitted coefficients were $\beta_0 = -1.72$, $\beta_1 = -0.073$ for distance, and $\beta_2 = 0.023$ for angle, all highly significant. As odds ratios, each extra unit of distance multiplies the odds of scoring by 0.93, a reduction of about 7 percent, and each extra degree of angle multiplies them by 1.02. The resulting xG surface is shown in Figure 3.

6 Evaluation

Models are assessed on a held-out test set of 20 percent of shots, using three measures. The area under the ROC curve (AUC) is the probability that a random goal is given a higher xG than a random non-goal, measuring ranking only. The Brier score, $BS = \frac{1}{n} \sum_i (p_i - y_i)^2$, measures the accuracy of the probabilities, with lower being better. Calibration is checked with a reliability plot and summarised by the expected calibration error (ECE), the average gap between predicted probability and observed rate.

The baseline reached an AUC of 0.746 and a Brier score of 0.0795, beating the no-skill reference of 0.086. Its reliability curve (Figure 4) sits almost exactly on the diagonal, so its probabilities can be read as genuine chances.

7 Contextual Features and the Enhanced Model

The descriptive columns were turned into numeric features by one-hot encoding, where each category becomes a 0/1 column. For each feature one level was left out as a reference (right foot, normal technique, open play, regular play), so every coefficient reads as a comparison to a typical shot. Figure 5 shows how conversion varies across these categories. Headers and shots from a counter attack convert relatively well, and first-time shots convert better than shots where the player takes a touch first.

Fitting the logistic model on all 20 features raised the AUC to 0.777 and improved the Brier score to 0.0772. The coefficients, shown as odds ratios in Figure 6, are revealing. Headers now

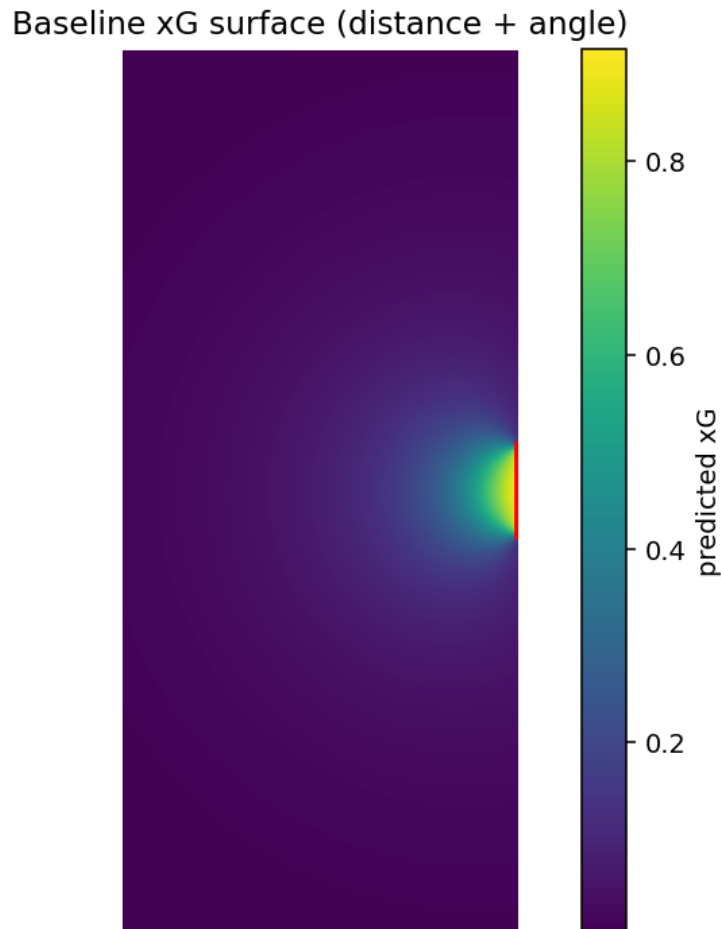


Figure 3: Baseline xG surface over the attacking third, brightest in front of goal.

have the strongest negative effect, an odds ratio near 0.22, which appears to contradict their high raw conversion. The explanation is that headers convert well only because they occur close to goal, and once distance and angle are already accounted for, a header from the same position is clearly worse than a foot shot. This is the model separating location from technique, and it illustrates why a multi-feature model gives a fairer picture. Shots under pressure score less often, and shots from a counter attack score more.

8 Tree Models and Comparison

Two tree based models were fitted with the same features, split, and evaluation. A Random Forest grows many decision trees on random slices of the data and averages them, while XGBoost builds trees one after another, each correcting the errors of the last. Both were tuned lightly using a validation slice of the training data.

Table 1 and Figure 7 tell a clear story. The largest single gain comes from adding contextual features to the baseline. After that, the enhanced logistic model, the Random Forest, and XGBoost are almost indistinguishable on every measure. This means the models have reached the ceiling of what the available features can explain, and the remaining gap to the StatsBomb benchmark is due to information, chiefly the positions of defenders and the goalkeeper, rather than to model choice. All models are well calibrated, and on calibration error the XGBoost model matches the professional benchmark. Applying Platt scaling or isotonic regression to the already-calibrated XGBoost model did not improve test-set calibration, so no correction was

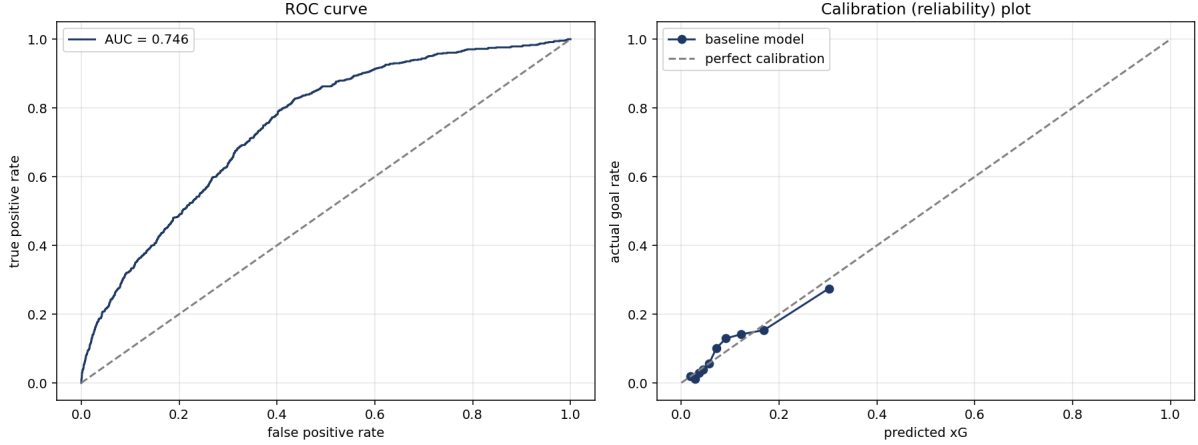


Figure 4: Baseline model: ROC curve (left) and calibration plot (right).

Table 1: Model comparison on the held-out test set. Arrows show the good direction.

Model	AUC $\uparrow$	Brier $\downarrow$	LogLoss $\downarrow$	ECE $\downarrow$
Baseline logit (distance + angle)	0.746	0.0795	0.280	0.0162
Enhanced logit (context)	0.777	0.0772	0.270	0.0138
Random Forest	0.779	0.0769	0.269	0.0126
XGBoost	0.780	0.0769	0.269	0.0114
StatsBomb xG (benchmark)	0.816	0.0714	0.251	0.0116

retained.

9 Applied Outputs

XGBoost was used as the final model for the applied outputs, being the strongest on every measure and the best calibrated. To avoid a model scoring shots it was trained on, each shot was assigned an xG using five-fold cross-validation. As a whole-dataset check, the total xG of 3,568 almost exactly matched the 3,569 actual goals, confirming good calibration across the entire sample.

Figure 8 maps shots coloured by xG, with the highest values concentrated centrally in the box. Finishing was then measured for each player as actual goals minus total xG. Figure 9 shows the biggest over and under-performers among players with at least 40 shots. The over-performers are led by elite forwards of the era such as Suárez, Ibrahimovic, Bale, Higuaín, and Griezmann, which is strong evidence that the model behaves sensibly. As the literature review noted, a single season of finishing is noisy and biased, so this table describes who out-performed their chances in this season rather than a permanent measure of skill.

10 Match Simulation

Once every shot has an xG value p_i , a shot can be treated as an independent Bernoulli trial. A team’s goals in a match, $G = \sum_i B_i$ with $B_i \sim \text{Bernoulli}(p_i)$, then follow a Poisson-binomial distribution with mean $\sum_i p_i$ and variance $\sum_i p_i(1 - p_i)$. Treating the two teams’ goals as independent, the match outcome probabilities are $P(\text{home win}) = P(G_H > G_A)$ and so on, and the expected points for the home team are

$$\mathbb{E}[\text{pts}_H] = 3P(G_H > G_A) + 1P(G_H = G_A). \quad (5)$$

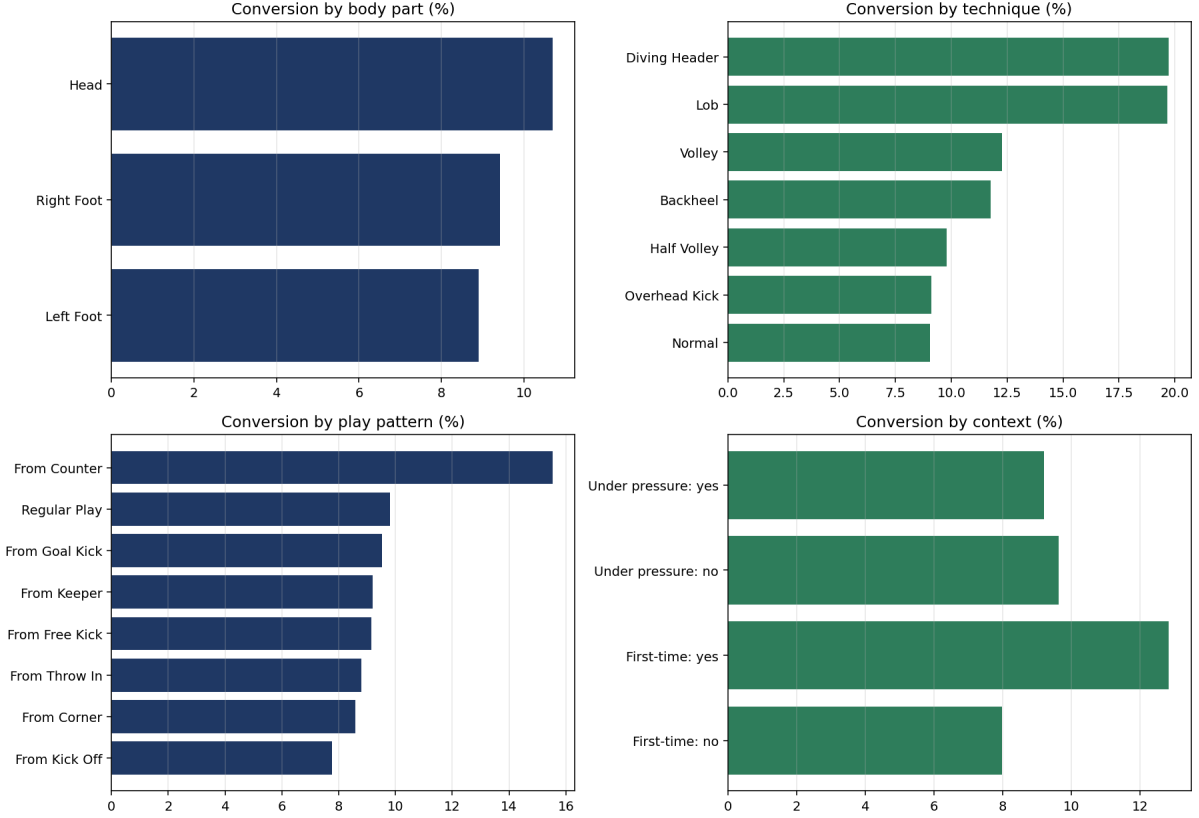


Figure 5: Empirical conversion rate by body part, technique, play pattern, and shot context.

These quantities were estimated by Monte Carlo, simulating every match 10,000 times. Figure 10 validates the method on the highest-shot-volume match: the simulated goal distribution matches the exact Poisson-binomial distribution, and the outcome probabilities show that Juventus deserved to win 71 percent of the time yet only drew.

Summing expected points across a season gives an xG-based league table. Figure 11 compares actual points to expected points for all four leagues. Teams above the diagonal earned more real points than their chances deserved. The two largest over-performers were Leicester City (+20) and Atlético Madrid (+21), which matches the well-known story of the 2015/16 season, when Leicester won the Premier League against expectation and Atlético were famed for their efficiency.

11 Bayesian Model and Uncertainty

The models above give a single estimate for each quantity. A Bayesian treatment produces a full posterior distribution and therefore credible intervals. Keeping the same likelihood and placing weakly-informative priors $\alpha \sim \mathcal{N}(0, 5^2)$ and $\beta_k \sim \mathcal{N}(0, 5^2)$, the posterior $p(\alpha, \beta \mid \mathbf{y})$ was sampled. Because the dataset is large, the Bernstein–von Mises theorem ensures the posterior is close to $\mathcal{N}(\hat{\beta}, \hat{\Sigma})$, which was used for the results here and reproduced with full Hamiltonian Monte Carlo in Stan.

The posterior for the distance coefficient had a 95 percent credible interval of about $[-0.085, -0.068]$ and the angle coefficient $[0.018, 0.024]$, both well away from zero. Figure 12 shows the coefficient posteriors and a credible band on the xG versus distance curve. The band is wider close to goal, where fewer shots are taken and the model is less certain, and narrower far out. This is uncertainty quantification made explicit: any prediction can now be reported with a credible interval.

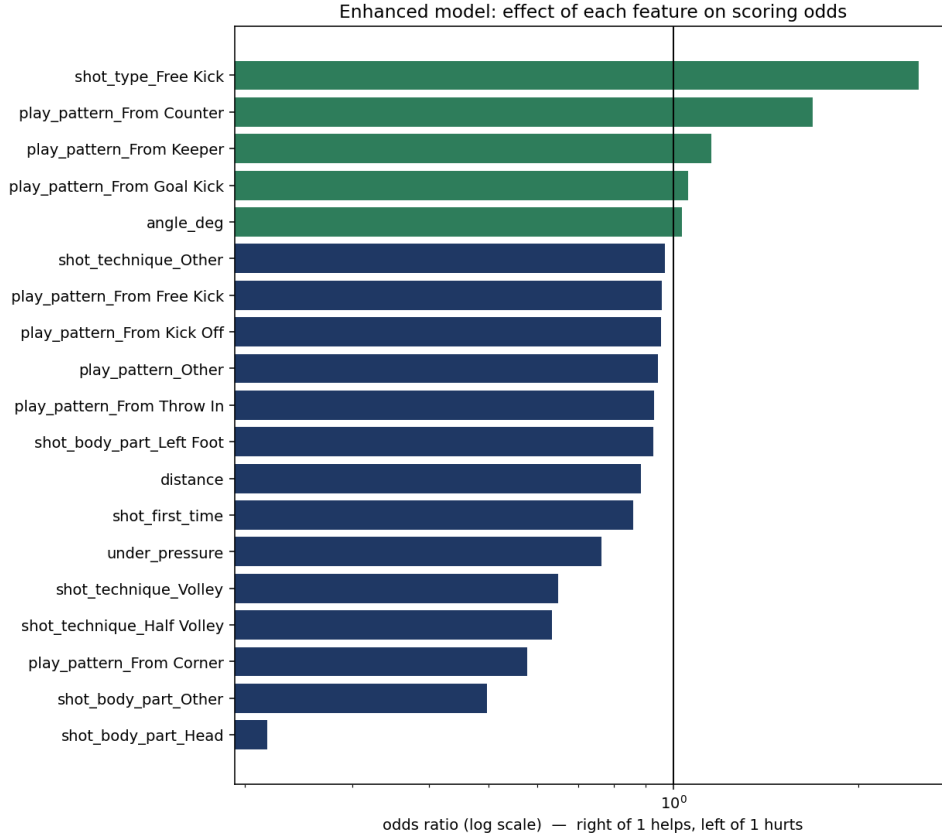


Figure 6: Effect of each feature in the enhanced model. Bars right of 1 raise the odds of scoring, bars left of 1 lower them.

12 Sensitivity Analysis

The behaviour of the model was studied both locally and globally. Locally, the derivatives of the logistic model follow from $\sigma'(\eta) = \sigma(\eta)(1 - \sigma(\eta))$:

$$\frac{\partial p}{\partial d} = \beta_1 p(1 - p), \quad \frac{\partial p}{\partial \theta} = \beta_2 p(1 - p). \quad (6)$$

Both share the factor $p(1 - p)$, which is largest when $p = 0.5$. The model is therefore most sensitive to position for shots of moderate xG, physically a ring around the edge of the six-yard box, and least sensitive for near-certain or near-hopeless chances. This is visible in Figure 13.

Globally, variance-based Sobol indices measure how much of the variation in xG across all shots is driven by each input. Distance had a first-order index of 0.62 and angle 0.31, so distance is the dominant driver with angle second. The small gap between the first-order and total-effect indices (0.62 against 0.69 for distance) indicates only a modest interaction between the two features (Figure 14).

13 Discussion and Limitations

The results support three main conclusions. First, most of the model's predictive power comes from a small set of geometric and contextual features, and the largest single improvement is the move from distance and angle alone to the full feature set. Second, once these features are fixed, flexible models offer almost no advantage over a well-specified logistic regression, so the interpretable model is a reasonable final choice. Third, the gap to a professional model is

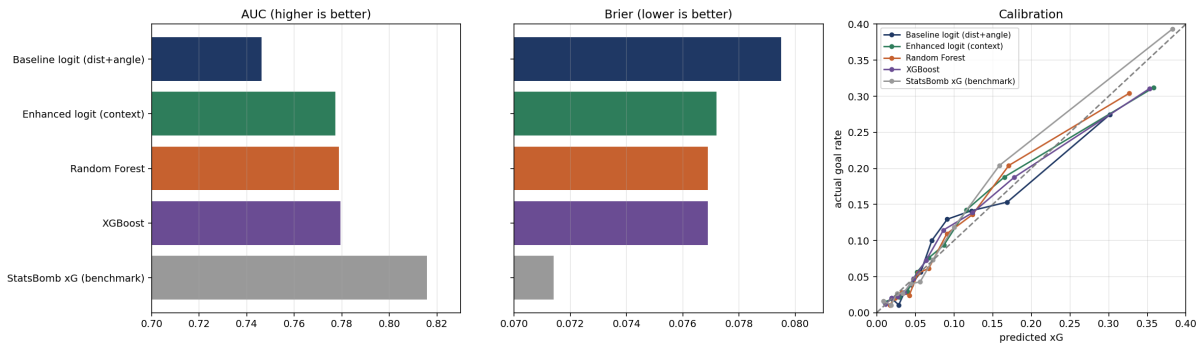


Figure 7: AUC, Brier score, and calibration across all models.

explained by missing information rather than by method, which is a useful and honest boundary on what open event data can achieve.

Several limitations should be noted. The data lack freeze-frame information on defender and goalkeeper positions, which is the single largest missing input and the main reason for the gap to the StatsBomb benchmark. Only one fully covered season was available, so the model cannot be tested across seasons. The finishing table is based on a single season and, as the literature shows, such measures are noisy and biased, so they should not be read as fixed skill. The match simulation treats the two teams' goals as independent and ignores game state, and it uses open-play, non-penalty shots only, so simulated scorelines omit penalties and own goals. Finally, the global sensitivity analysis assumes distance and angle vary independently and uniformly, whereas in real shots they are correlated.

14 Conclusion

This project built a calibrated xG model on open data and treated it as a mathematical model throughout. The shot geometry was derived from first principles, the baseline fit was framed as a maximum likelihood optimisation problem, and the model was extended to match outcomes through a Poisson-binomial Monte Carlo simulation, to uncertainty through a Bayesian treatment, and to behaviour through a sensitivity analysis. The final model is accurate, well calibrated, and interpretable, and its outputs recover known real-world results. The clearest route to further improvement would be richer positional data, which the analysis identifies as the main factor separating this model from a professional one.

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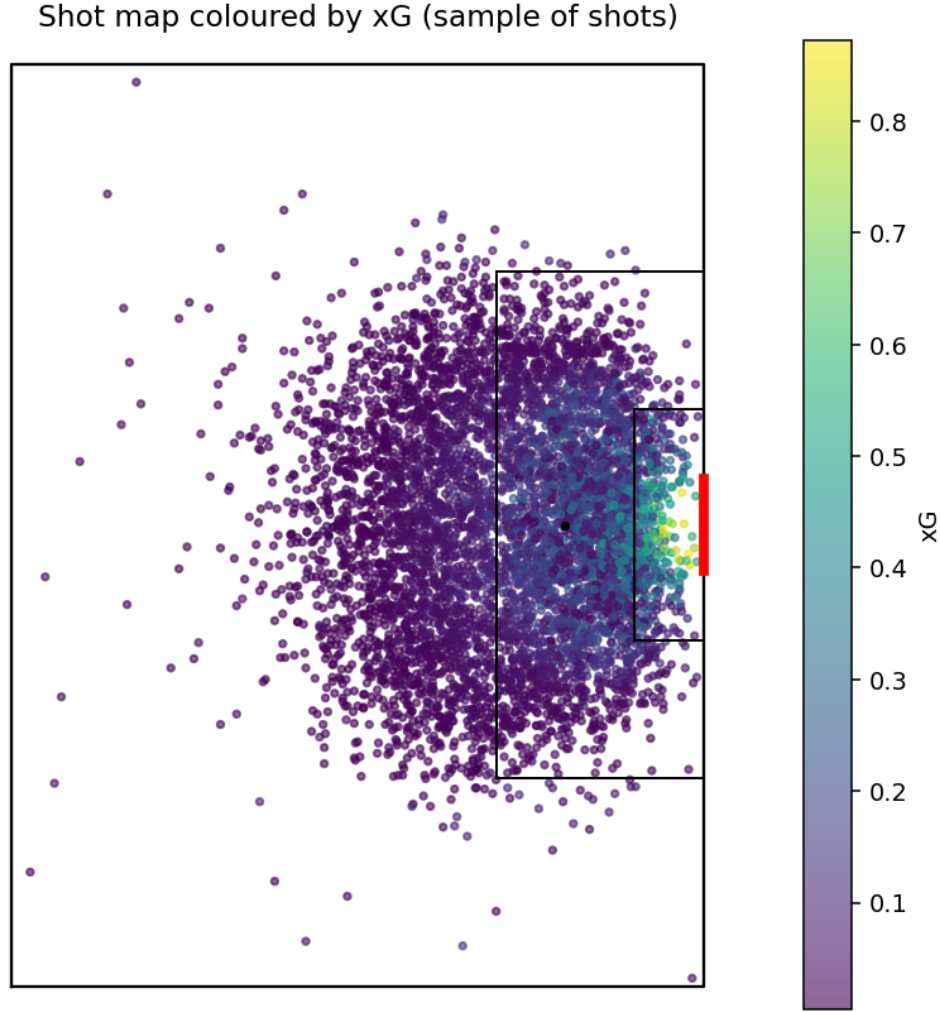


Figure 8: A sample of shots coloured by predicted xG.

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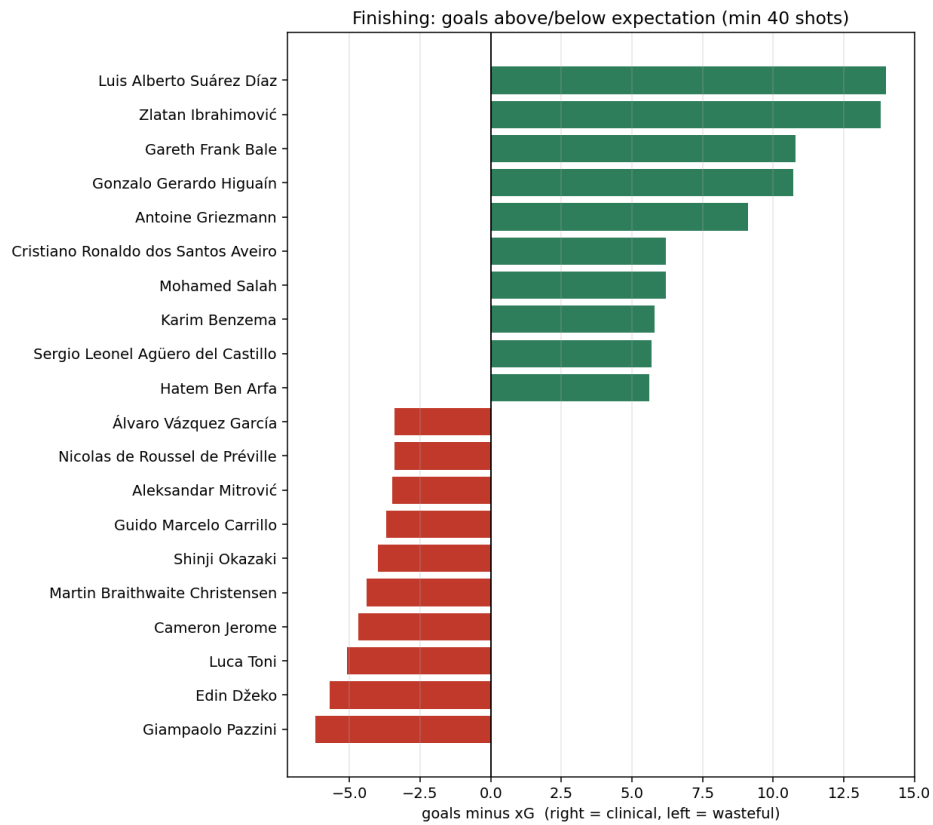


Figure 9: Finishing: goals above or below expectation, players with at least 40 shots.

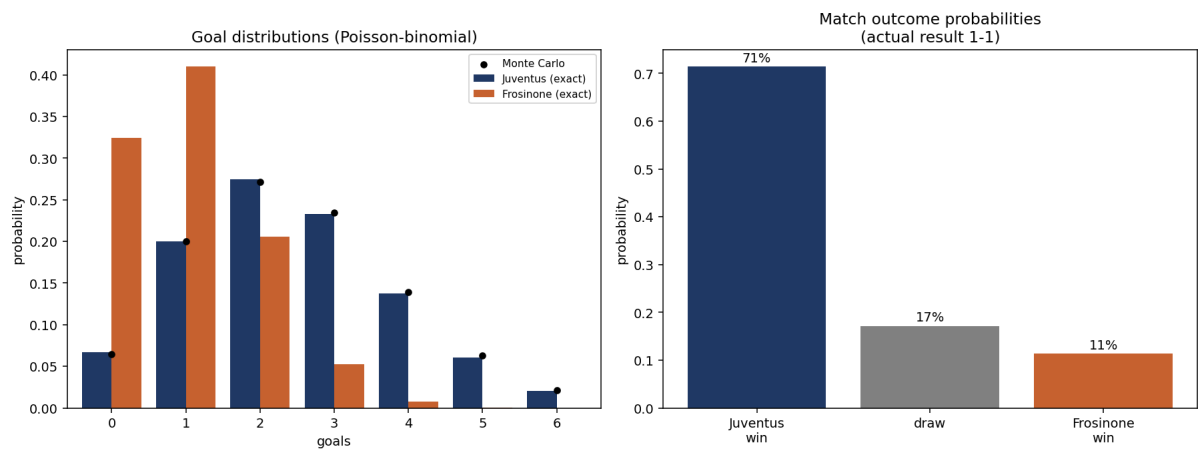


Figure 10: Example match. Left: Monte Carlo goal counts (dots) against the exact Poisson-binomial distribution (bars). Right: outcome probabilities.

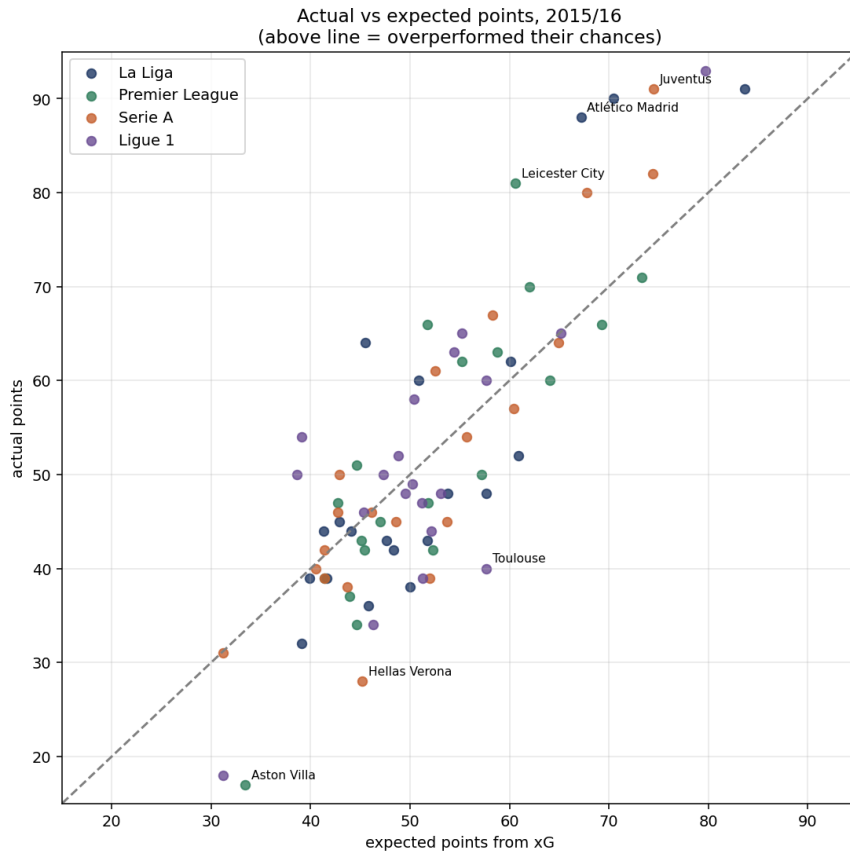


Figure 11: Actual points against expected points from xG. Teams above the line over-performed their chances.

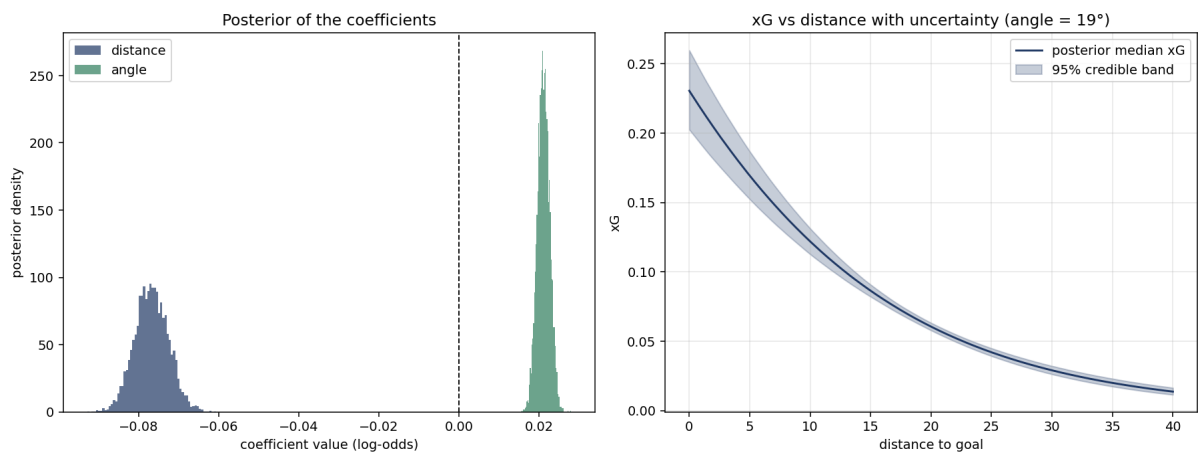


Figure 12: Posterior distributions of the coefficients (left) and xG against distance with a 95 percent credible band (right).

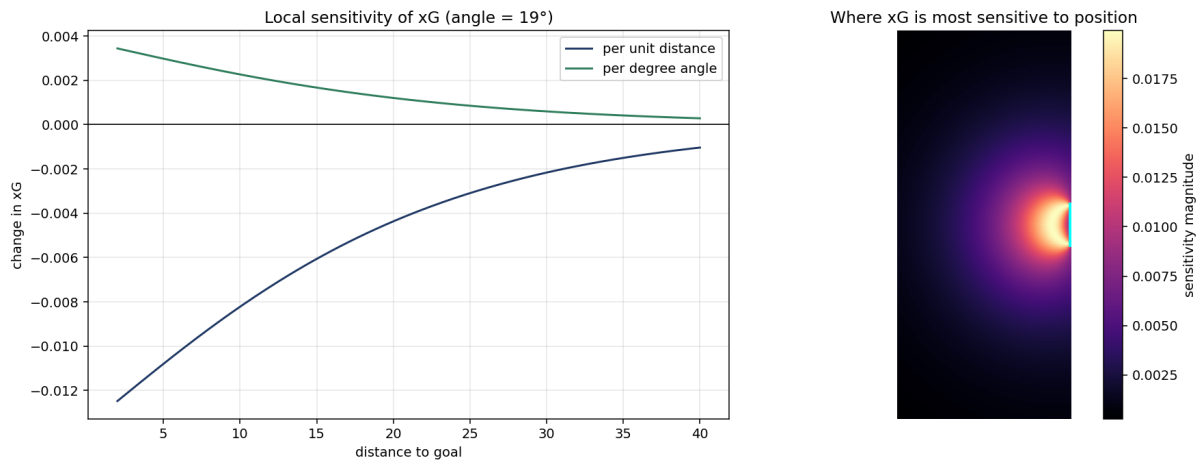


Figure 13: Local sensitivity: change in xG per unit distance and per degree of angle (left), and where on the pitch xG is most sensitive to position (right).

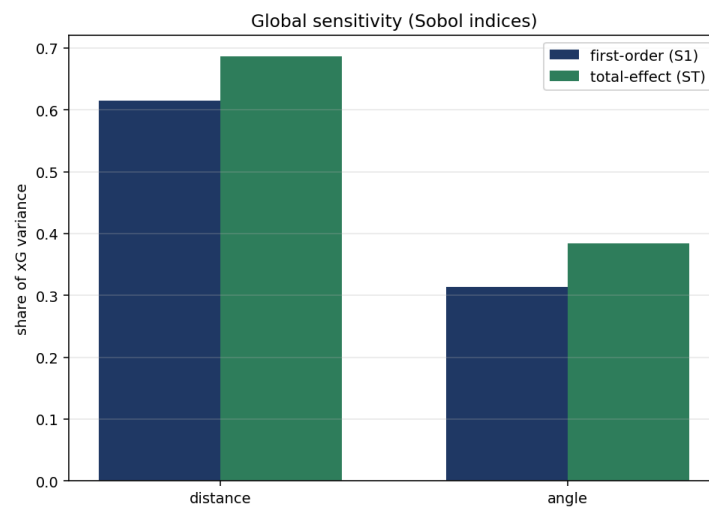


Figure 14: Global sensitivity: first-order and total-effect Sobol indices.